Lecture 6

- Positive Definite Matrices
- Gram Matrices

Last time, in addition to the standard dot product, we defined the weighted inner product on $\mathbb{R}^n$:

$$\langle v, w\rangle = \sum_{i=1}^{n} \alpha_i v_i w_i \quad , \text{ where } \alpha_i > 0 .$$

This generalization of the dot product leads to the following natural question: <u>what is the most general inner product that can be defined on $\mathbb{R}^n$?</u>

Let $\langle\cdot,\cdot\rangle$ be an inner product on $\mathbb{R}^n$, and let $x, y \in \mathbb{R}^n$ be any two vectors

$$\langle x,y\rangle = \langle \sum_{i=1}^{n} x_i e_i, \sum_{j=1}^{n} y_j e_j\rangle = \sum_{i,j=1}^{n} x_i y_j \underbrace{\langle e_i, e_j\rangle}_{k_{ij}} = \sum_{i,j=1}^{n} k_{ij} x_i y_j = x^T K y .$$

Thus, any inner product can be written as

$$\langle x,y\rangle = x^T K y \qquad \text{where } K = [k_{ij}], \quad k_{ij} = \langle e_i, e_j\rangle$$

What can we say about matrix K? Can it be arbitrary?

- Since $\langle\cdot,\cdot\rangle$ is symmetric: $k_{ij} = \langle e_i, e_j\rangle = \langle e_j, e_i\rangle = k_{ji} \Rightarrow$ <u>K is symmetric</u> $K^T = K$.
- Since $\langle\cdot,\cdot\rangle$ is positive definite: $\langle x,x\rangle \geq 0$ and $\langle x,x\rangle = 0$ iff $x=0$

$$\Rightarrow \quad \langle x,x\rangle = x^T K x \geq 0 \quad \text{and} \quad x^T K x = 0 \iff x = 0 .$$

It is not immediately clear what this condition means and how it affects properties of matrix K, so it deserves a separate definition.

Def A symmetric matrix K is called <u>positive definite</u> (denoted $K>0$) if $x^T K x > 0 \quad \forall x \in \mathbb{R}^n, x \neq 0$

Remark 1 $K>0$ does not mean $k_{ij}>0$.

For example, $K = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ is positive definite: $(x_1\ x_2)\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1\ x_2)\begin{pmatrix} 2x_1 - x_2 \\ 2x_2 - x_1 \end{pmatrix}$

$$= 2x_1^2 - x_1x_2 + 2x_2^2 - x_1x_2 = 2(x_1^2 - x_1x_2 + x_2^2) > 0$$

Remark 2 $k_{ij}>0$ does not imply $K>0$

For example, $K = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is not positive definite: $(x_1\ x_2)\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1 + x_2) = 0$ if $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$K>0 \underset{\not\Leftarrow}{\not\Rightarrow} k_{ij}>0$$

To each square matrix A, we can associate its

- bilinear form $b_A(x,y) = x^T A y$, and
- quadratic form $q_A(x) = x^T A x$

We have shown above that any inner product on $\mathbb{R}^n$ can be written as a bilinear form of a symmetric positive definite matrix.

It turns out that the converse is also true. Namely, if $K^T = K$ and $K > 0 \Rightarrow b_K(x,y)$ defines an inner product on $\mathbb{R}^n$

- $b_K(x,y)$ is bilinear by definition for any K.
- Symmetry: $b_K(x,y) = x^T K y \overset{a^T = a \text{ for any number}}{=} (x^T K y)^T = y^T K^T x \overset{K^T = K}{=} y^T K x = b_K(y,x)$
- Positive definite: $b_K(x,x) = x^T K x > 0 \quad \forall x \neq 0$ since $K > 0$.

Let us formulate the obtained result as theorem.

Th 1 A function $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defines an inner product on $\mathbb{R}^n$ if and only if it can be written as a bilinear form associated with symmetric positive definite matrix: $\boxed{\langle x,y \rangle = x^T K y}$, $K^T = K$, $K > 0$

Remark The dot product corresponds to $K = I_n$

The weighted inner product corresponds to $K = \begin{pmatrix} \alpha_1 & & 0 \\ & \ddots & \\ 0 & & \alpha_n \end{pmatrix}$, $\alpha_i > 0$

Thus, inner products on $\mathbb{R}^n$ are fully described by symmetric positive def. matrices.

One obvious property: If $K > 0 \Rightarrow \underbrace{\ker K = \{0\} \Leftrightarrow K \text{ is nonsingular}}_{\text{this is necessary condition, but not sufficient}}$

Gram Matrices

Def: Let V be an inner space and let $v_1 \dots v_n \in V$.

The associated Gram matrix is $K = [k_{ij}]$, $k_{ij} = \langle v_i, v_j \rangle$

Example Above we had $k_{ij} = \langle e_i, e_j \rangle$, where $\{e_i\}$ is the standard basis in $\mathbb{R}^n$

- All Gram matrices are symmetric: $k_{ij} = \langle v_i, v_j \rangle = \langle v_j, v_i \rangle = k_{ji}$
- Are Gram matrices positive definite? Not exactly, they are positive semi-definite

$q_K(x) = x^T K x \geq 0 \quad \forall x \in \mathbb{R}^n$ (denoted $K \geq 0$)

Th 2 If K is a Gram matrix $\Rightarrow K \geq 0$.

A Gram matrix $K > 0 \Leftrightarrow v_1 \dots v_n$ are linearly independent

Proof Let K be a Gram matrix associated with $v_1, \dots, v_n$

$$q_K(x) = x^T K x = \sum_{i,j=1}^n k_{ij} x_i x_j = \sum_{i,j=1}^n \langle v_i, v_j \rangle x_i x_j =$$

$$= \Big\langle \underbrace{\sum_{i=1}^n x_i v_i}_{u}, \underbrace{\sum_{j=1}^n x_j v_j}_{u} \Big\rangle = \langle u, u \rangle = \|u\|^2 \geq 0 \text{ for any } x \in \mathbb{R}^n \Rightarrow K \geq 0$$

$K > 0 \Leftrightarrow \forall x \in \mathbb{R}^n, x \neq 0 \ \|u\| > 0 \Leftrightarrow \forall x \in \mathbb{R}^n, x \neq 0 \ u = x_1 v_1 + \dots + x_n v_n \neq 0 \Leftrightarrow v_1 \dots v_n$ are lin. indep.

(*) → page 31

$\square$

Gram matrices can be associated with sets of vectors of any, even infinite dimensional inner space V.

<u>Example</u>: Let $V = C[0,1]$ equipped with the L^2 inner product $\langle f,g\rangle = \int_0^1 f(x)g(x)\,dx$. Let us construct the Gram matrix H_n associated with monomials $1, x, x^2, \dots, x^{n-1}$.

$$k_{ij} = \langle x^{i-1}, x^{j-1}\rangle = \int_0^1 x^{i+j-2}\,dx = \frac{1}{i+j-1}, \quad \begin{matrix} i = 1\dots n \\ j = 1\dots n \end{matrix}$$

$$\Rightarrow H_n = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots & \frac{1}{n+2} \\ \vdots & & & & \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n-1} \end{bmatrix}$$

This is the $n\times n$ <u>Hilbert matrix</u>. It is a famos example of ill-conditioned matrix. "almost" singular

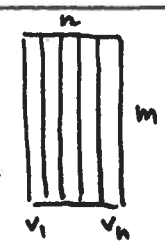
Since $1, x, \dots, x^{n-1}$ are linearly independent, $H_n > 0$. In particular, H_n is nonsingular for any n. But when n increases, H_n becomes closer to being singular (check the last row for instance). Even for moderately large n, solving $H_n x = b$ is a challenging problem. We will come back to H_n when we will be discussing the <u>condition number</u> of a matrix

Let us now go back to our "pet" vector space $V = \mathbb{R}^m$.

<u>Q</u>: How does a general Gram matrix G associated with $v_1 \dots v_n \in \mathbb{R}^m$ look like?

$$g_{ij} = \langle v_i, v_j\rangle \overset{\text{Th1}}{=} v_i^T K v_j, \quad \text{for some } K^T = K,\ K > 0$$

Let us rewrite this in a matrix form.

Let A be an $m\times n$ matrix whose columns are $v_1 \dots v_n$: $A =$ 

$$\Rightarrow \boxed{G = A^T K A}$$

defines $\langle\cdot,\cdot\rangle$ — defines $v_1 \dots v_n$

<u>Special case</u>: If $\langle\cdot,\cdot\rangle$ is the dot product $\Rightarrow G = A^T A$

This factorization allows to establish an important property of positive definite matrices.

<u>Th3</u> If $K > 0$ and rank $A = n \Rightarrow G > 0$ $\left(\begin{matrix} G = A^T K A \\ A \text{ is } m\times n \end{matrix}\right)$

<u>Proof</u> $G \geq 0$ since G is a Gram matrix. (associated with columns of A) Th2

If rank $A = n \Rightarrow$ columns of A are linearly independent $\Rightarrow G > 0$ $\square$

Gram matrices $G = A^T K A$ arise in a wide variety of applications, including least squares approximations, which we will discuss next time.

Every inner product on $\mathbb{R}^n$ induces a symmetric positive definite matrix. Let us address the inverse problem: given $A \in \mathbb{M}_{n\times n}$, when does it induces an inner product on $\mathbb{R}^n$? Obviously, when $A^T = A$ and $A > 0$. Checking $A^T = A$ is easy, but how to check whether $A > 0$ or not? So far we know only a necessary condition: $\det A \neq 0$. Before we formulate the criterion let us develop a low-dimensional intuition for positive definiteness.

$n=1$

- Let $A = a \in \mathbb{M}_{1\times 1}$. Its positive def. $\iff$ $x^T a x = a x^2 > 0 \quad \forall x \neq 0$

  Thus, for a scalar a, positive def. simply means positivity: $a > 0$.

$n=2$

- Let $A \in \mathbb{M}_{2\times 2}$ be a symmetric matrix: $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

$$q_A(x) = (x_1, x_2)\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1, x_2)\begin{pmatrix} a x_1 + b x_2 \\ b x_1 + c x_2 \end{pmatrix} = a x_1^2 + 2b x_1 x_2 + c x_2^2$$

$$q_A(x) > 0 \quad \forall x \neq 0 \iff \begin{cases} a > 0 \\ \underbrace{x_1^2 + \frac{2bx_2}{a} x_1 + \frac{c x_2^2}{a} > 0}_{\Updownarrow} \quad \forall x_2 \neq 0 \end{cases}$$

discriminant

$$D = \left(\frac{b x_2}{a}\right)^2 - \frac{c x_2^2}{a} < 0 \quad \forall x_2 \neq 0$$

or, equivalently: $ac - b^2 > 0$.

Thus, A is pos. def. if and only if

$$\begin{cases} a > 0 \\ \underbrace{ac - b^2}_{\det A} > 0 \end{cases}$$

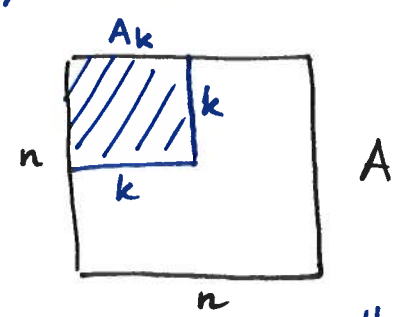
In general case, we have the following criterion.

<u>Sylvester's Criterion</u>: A symmetric matrix A is positive definite $\iff$ all its leading principal minors are positive.

No need to consider this generalization

<u>Remark</u> The symmetry condition is important.

(skip) $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ All principal minors are positive, but $(1,-1)\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = (1\ -1)\begin{pmatrix} -1 \\ -1 \end{pmatrix} = 0$



$\det A_k$ is the kth leading principal minor of A

This leads to a natural question:

How to check whether $A > 0$ or not if $A \neq A^T$?

Here is a trick:

$$A > 0 \iff \forall x \neq 0 \quad x^T A x > 0 \iff \forall x \neq 0 \quad \underbrace{\overbrace{2}^{\text{number}} x^T A x}_{} > 0$$

$$= x^T A x + (x^T A x)^T = x^T A x + x^T A^T x = x^T (A + A^T) x$$

Thus $A > 0 \iff \underbrace{A + A^T}_{\text{symmetric}} > 0$ $\leftarrow$ we can use Sylvester's Criterion.

(**)

(*) Positive semi-definite matrices is a more general class of matrices than positive definite matrices: $\{A : A>0\} \subset \{A : A \geq 0\}$

Def A vector x is a null direction for matrix A if $q_A(x) = x^T A x = 0$.

- If $K>0 \Rightarrow K$ has only trivial null direction $x=0$
- If $K \geq 0 \Rightarrow K$ may have non trivial null directions $x \neq 0$.

It is clear that if $x \in \ker A \Rightarrow x$ is a null direction of A.

In general, the converse is not true: However, if $\overset{K^T=K}{K \geq 0}$, then $\left\{\begin{matrix}\text{null}\\ \text{directions}\end{matrix}\right\} = \ker K$

Example: Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and let $x = (1,1)$. Then:

- $x^T A x = (1,1)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = (1,1)\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$
- $Ax = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

x is a null direction, but $x \notin \ker A$

Th 2.5 Let $K \geq 0$ and $K^T = K$. Then

$$x^T K x = 0 \iff Kx = 0$$

Proof ($\Leftarrow$) Is obvious. (Skip the proof in class.)

($\Rightarrow$) Suppose that $x \neq 0$ and $x^T K x = 0$.

$K \geq 0$ let $y = \alpha x + Kx$. Let us compute $q_K(y)$.

$$0 \leq q_K(y) = (\alpha x + Kx)^T K (\alpha x + Kx) =$$

$$= \alpha^2 \underbrace{x^T K x}_{0} + \alpha \underbrace{x^T K^T}_{(K\cdot x)^T} K x + \alpha \underbrace{x^T K}_{(Kx)^T} \cdot Kx + x^T K^T K K x = 2\alpha \|Kx\|^2 + x^T K^3 x$$

since $K^T = K$.

If $Kx \neq 0 \Rightarrow \|Kx\| \neq 0$ and we can take α sufficiently large and negative so that $q_K(y) < 0$.

$\Rightarrow Kx = 0$. □

Remark: On p. 156 of Olver & Shakiban, it is written that there may be null directions other than $\ker K$. This is not true in view of Th. 2.5

Corollary A symmetric positive semi-definite matrix ($K \geq 0$) is positive definite ($K>0$) if and only if it is nonsingular ($\ker K = 0$).

(**) If n is large, the Sylvester's Criterion is not efficient (computing det. of large matrices is comp. expensive)

A more efficient way is to use the Cholesky decomposition

If $K>0$ and $K^T=K \Rightarrow K = R^T R$, where R is an upper triangular matrix with positive diag. elements

In Matlab, there is function chol that computes the Cholesky decomposition.

$R = \begin{bmatrix} + & * \\ 0 & + \end{bmatrix}$

It can also be used for checking positive def:

$[\sim, p] = \text{chol}(A)$ | $A>0 \Rightarrow p=0$; otherwise $p>0$

Remark: this is more efficient than computing eigenvalues

Remark: chol(A) expects symmetric matrix